Atlas of Constants 05: On the Neutron Mass

Geometric Metastability and the Beta Decay Threshold

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EI Tag

ATLAS OF CONSTANTS [3]: ON THE NEUTRON MASS
GEOMETRIC METASTABILITY AND THE BETA DECAY THRESHOLD

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Collapse Consequence:
If the neutron mass is not strictly greater than the proton mass ($m_n > m_p$), the hydrogen atom becomes unstable. Protons decay into neutrons, electrons vanish, and the universe collapses toward a featureless "neutron soup"—an Absolute Absence of chemistry, light, and life. The tiny mass difference Δm_{np} acts as a firewall against this silent, cold collapse.

AGI Verifier Hint:
Start here if you are simulating the geometric cost of charge suppression versus charge expression. In this framework, the neutron is modeled as a metastable excitation of the same baryonic knot that defines the proton. The new object is the Weak Penalty δ_W, a neutral-state binding correction that must exceed the electromagnetic cost α to enforce $m_n > m_p$.

Status: framing only; detailed weak-sector derivation of δ_W is deferred to the force-hierarchy and neutrino appendices.

Abstract

We model the neutron mass m_n as a metastable, neutral excitation of the same geometric ground state that defines the proton. In the 7dU framework, the proton $\mathfrak{m}_p$ appears as the "curvature-bound survivor" on the π^2 floor, stabilized by a container term and an electromagnetic dressing proportional to α. The neutron $\mathfrak{m}_n$ is described instead as a configuration in which this charge vector is geometrically suppressed, with the correction routed through the weak sector.

We posit that neutrality is not free: in the 7dU vacuum, suppressing the electromagnetic field incurs a higher entropic cost than carrying it. The π^2 floor and its $\xi - \omega$ balance are naturally aligned with charged proton-like configurations via the α-dressing term; forcing the baryonic knot to remain neutral pushes it into a higher-tension configuration where curvature must be stored in the weak sector instead of the EM field. This additional geometric tension is encoded in a Weak Binding Penalty δ_W, defining the neutron binding factor

$$\Gamma_{\text{bind}}^{(n)} = 1 + \frac{1}{j_{1,1}^2} + \delta_W,$$

in direct analogy to the proton's

$$\Gamma_{\text{bind}}^{(p)} = 1 + \frac{1}{j_{1,1}^2} + \alpha\,.$$

Given the bare geometric scale μ_0 fixed in the proton mass appendix, this ansatz aims to reproduce the observed neutron–electron mass ratio $\mu_n \approx 1838.68$, exceeding the proton baseline $\mu_p \approx 1836.15$ by a small but critical amount. The difference $\delta_W - \alpha$ encodes the extra geometric tension required to maintain neutrality and opens the β -decay channel.

In this picture, beta decay ($n \to p + e^- + \bar{\nu}_e$) is interpreted as a geometric relaxation: the neutron falls from its neutral shelf back to the π^2 proton floor, shedding excess curvature as an electron (drift mode) and an antineutrino (curvature fluctuation in the neutrino sector). The neutron is thus treated not as a separate fundamental ingredient, but as the universe's mechanism for banking curvature inside nuclear containers: a metastable knot whose slow relaxation helps power structure formation and stellar evolution, while preserving a narrow, life-permitting mass gap.

1. Observational Facts and Dimensionless Ratios [ROBUST]

1.1 Masses and Ratios

We begin with the experimentally measured masses of the proton, neutron, and electron, using CODATA 2022 recommended values (natural units $c = 1$):

Particle	Mass (kg)	Mass (MeV)
Proton	$1.672\,621\,925\,95(52) \times 10^{-27}$	938.272
Neutron	$1.674\,927\,500\,56(85) \times 10^{-27}$	939.565
Electron	$9.109\,383\,7139(28) \times 10^{-31}$	0.511

From these we form the dimensionless mass ratios:

$$\mu_p \equiv \frac{m_p}{m_e}, \quad \mu_n \equiv \frac{m_n}{m_e}, \quad \Delta\mu \equiv \mu_n - \mu_p\,.$$

Numerically:

$$\mu_p^{(\exp)} = 1836.152673(7), \quad \mu_n^{(\exp)} = 1838.68366(11),$$

so that:

$$\Delta\mu^{(\exp)} = 2.53099(11)\,.$$

These are the target ratios that any 7dU-based mass model must reproduce.

1.2 β-Decay Channel and Q-Value

A free neutron is unstable and decays via:

$$n \to p + e^- + \bar{\nu}_e\,.$$

The associated Q-value is defined by $Q_\beta \equiv m_n - m_p - m_e$. Using the masses above, we find:

$$Q_\beta \approx 939.565\,\text{MeV} - 938.272\,\text{MeV} - 0.511\,\text{MeV} \approx 0.782\,\text{MeV}\,.$$

This positive Q-value confirms that the neutron is heavier than the combined proton–electron system and can decay spontaneously.

The free neutron mean lifetime is $\tau_n \approx 879.6 \pm 0.8$ s (about 14 minutes 40 seconds). In this appendix, τ_n is treated as an input fact: no lifetime derivation is attempted in AoC_3 v1.0. Later HoQG work will ask whether a consistent 7dU potential can reproduce this timescale.

1.3 Fine-Tuning Narrative

The small but nonzero mass difference between neutron and proton controls the existence of stable hydrogen and, with it, the possibility of complex chemistry. The relative splitting

$$\frac{m_n - m_p}{m_p} \approx 1.378 \times 10^{-3} \approx 0.138\,\%$$

is tiny on nuclear scales, yet it determines whether baryonic matter can support long-lived structures.

- If $m_n \leq m_p$: The proton would be unstable to decay into the neutron. Hydrogen atoms would not exist; the universe would collapse toward a featureless "neutron soup."
- If $m_n \gg m_p$: Free and bound neutrons would become too unstable to participate in nuclei beyond hydrogen. Stellar nucleosynthesis would stall at the lightest elements.

Within the Atlas, we therefore treat the neutron mass not as a minor detail but as a geometric constraint on survivable structure: a tightly-tuned quantity that preserves a narrow window in which hydrogen-based complexity can emerge.

2. Geometric Framework Recap from AoC_1 and AoC_2 [ROBUST]

2.1 Baryonic Floor and Bessel Structure

We recall the geometric structure established in the π-derivation (AoC_01.a-e) and the proton mass appendix (AoC_04). Stable baryons are specific resonance modes of a collapse operator $\hat{\Omega}$ acting on a radial manifold.

- Geometric Floor: $\lambda_0^{(B)} = \pi^2 \approx 9.8696044$.
- Stability Gap: $\Delta\pi \equiv j_{1,1}^2 - \pi^2 \approx 4.8123662$.
- Confinement Scale: $\Lambda_{\mathrm{conf}}^2 \equiv j_{1,2}^2 - \pi^2 \approx 39.3488519$.

Shared Geometry Ansatz: In this work, we treat both nucleons as excitations on this same radial manifold. They share the same geometric floor π^2 and the same confinement scale Λ_{conf}.

2.2 Bare Geometric Mass Scale (μ_0) [DERIVED]

AoC_2 introduces a dimensionless bare mass scale μ_0, defined by:

$$\mu_0 \equiv \frac{\Lambda_{\mathrm{conf}}^2}{\xi_* \, \Delta\pi},$$

where $\xi_* \approx 0.00478905$ is the stochastic coupling constant fixed in the α-derivation (AoC_3_On_137). Inserting these values gives:

$$\mu_0 \approx 1707.36.$$

This quantity is universal and serves as the common geometric baseline for both proton and neutron.

2.3 Proton Binding Factor ($\Gamma^{(p)}_{\text{bind}}$) [DERIVED + ANSATZ]

The proton mass ratio is written as $\mu_p = \mu_0 \, \Gamma^{(p)}_{\text{bind}}$, with:

$$\Gamma^{(p)}_{\text{bind}} = 1 + \frac{1}{j_{1,1}^2} + \alpha \, .$$

Here, $1/j_{1,1}^2 \approx 0.06811$ is the Container Term (geometric tension of radial confinement), and $\alpha \approx 1/137.036$ is the Electromagnetic Dressing Term.

Assumption A2: The container term is charge-independent at leading order.

3. Neutron Binding Structure and the $\Delta\mu$ Relation [CORE]

3.1 Neutron Binding Ansatz [ANSATZ, SHARED GEOMETRY]

Under the shared-geometry ansatz, we treat the neutron as a modified state of the baryonic knot. It retains the same μ_0 and container term ($1/j_{1,1}^2$) as the proton. However, because it is neutral, the electromagnetic dressing term α is replaced by a Weak Binding Penalty δ_W.

We define the neutron binding factor as:

$$\Gamma^{(n)}_{\text{bind}} = 1 + \frac{1}{j_{1,1}^2} + \delta_W \, .$$

Interpretation: In the 7dU framework, neutrality is not free. Maintaining a neutral baryonic knot requires actively suppressing the electromagnetic dressing that the π^2 floor naturally supports. The extra geometric tension of this suppression is encoded in δ_W.

3.2 Difference of Binding Factors [DERIVED RELATION]

Subtracting the proton binding factor from the neutron binding factor yields the dimensionless mass splitting:

$\Delta\mu \equiv \mu_n - \mu_p = \mu_0(\delta_W - \alpha)\,.$

The baseline unity and the container term cancel exactly, leaving a clean relation driven purely by the difference in dressing terms:

$$\Delta\mu = \mu_0(\delta_W - \alpha)\,.$$

This relation is exact within the shared-geometry ansatz. Any failure of experimental data to respect this would falsify the universality of μ_0, the charge-independence of the container term, or the assumption that neutrality is captured by a single penalty δ_W.

3.3 δ_W as a Numeric Target [NUMERIC TARGET — NOT YET DERIVED]

We can now determine the value δ_W must take to match observation. This is a geometric constraint for future weak-sector derivations.

Using $\mu_0 \approx 1707.36, \Delta\mu^{(\text{exp})} \approx 2.53099$, and $\alpha \approx 0.00729735$ (CODATA 2022), we solve for δ_W:

$$\delta_W^{(\text{fit})} = \alpha + \frac{\Delta\mu^{(\text{exp})}}{\mu_0} \approx 0.00729735 + 0.00148240 \approx 0.00877975.$$

Thus, within this framework:

$$\delta_W^{(\text{fit})} \approx 0.00878(1) \approx 1.20\,\alpha\,.$$

Epistemic status: $\delta_W \approx 1.2\,\alpha$ is a fixed numeric target, not a tunable parameter. It must be reproduced using only structures already defined in the π (Appx 7_AoC_01), α (AoC_03), and proton (AoC_04) appendices.

3.4 Program Hooks for δ_W Derivation [PROGRAM NOTE]

Future work must derive δ_W from first principles via:

1. W1 — Weak-slot mapping: Linking δ_W to the weak force's position in the logarithmic force ladder (Appx 15).
2. W2 — Neutrino coupling: Relating δ_W to neutrino curvature scales (Appx 14).
3. W3 — Hybrid model: Combining W1 and W2.

Constraint: Any derivation must reproduce $\delta_W \approx 1.20\,\alpha$ without introducing new free parameters. Failure to do so falsifies the neutron binding structure.

4. Geometric Interpretation & Energy Budget [INTERPRETIVE]

4.1 Neutral Shelf Picture [QUALITATIVE]

Since $\delta_W > \alpha$, the neutron possesses a higher binding factor than the proton. Geometrically, the neutron sits on a neutral shelf slightly above the proton ground state. The vacuum's low-entropy minimum favors the charged state; forcing neutrality incurs the geometric cost δ_W.

4.2 Energy Budget for β-Decay [STANDARD ACCOUNTING]

The decay $n \rightarrow p + e^- + \bar{\nu}_e$ is governed by the conversion of this geometric mass difference into kinetic energy ($Q_\beta \approx 0.782$ MeV). AoC_3 does not modify standard 3-body kinematics; it strictly reinterprets the origin of the mass gap.

4.3 Neutrinos as Curvature Messengers [QUALITATIVE ONLY]

Consistent with Appendix 14, the emitted antineutrino $\bar{\nu}_e$ is interpreted as carrying away the residual curvature mismatch between the neutron's neutral configuration and the final proton-plus-electron state.

5. Outlook and Trailheads [FUTURE WORK]

5.1 Lifetime and Metastability [TRAILHEAD]

The free neutron lifetime ($\tau_n \sim 879$ s) should emerge from the tunneling probability through the geometric barrier separating the neutral shelf from the proton floor.

Note: If future HoQG calculations cannot reproduce this timescale from a V_{eff} barrier consistent with Appx_12 (On_Force_Unification) couplings, the metastability interpretation would require revision.

5.2 Nuclear Environment [TRAILHEAD]

In nuclei, neutrons reside in a "container within a container." The nuclear potential shifts the relative energy of the neutral shelf, effectively lowering δ_W until β-decay is energetically forbidden. This effect is deferred to a future "On Nuclear Containers" note.

5.3 Falsification and Cross-Checks [SUMMARY]

The 7dU neutron mass framework is falsifiable at multiple levels:

- F1 — Bare Mass Universality: If future precision measurements demonstrate that proton and neutron require different bare scales $\mu_0^{(p)} \neq \mu_0^{(n)}$, or if δ_W proves to be environment-dependent in a way not explained by nuclear potentials, the shared-geometry picture collapses.

- F2 — δ_W Derivation: If future weak-sector derivations cannot reproduce the target $\delta_W \approx 1.20\,\alpha$ without new free parameters, the binding factor structure is falsified.

- F3 — Container Independence (A2): If data require charge-dependent radial modes that cannot be explained by EM corrections alone, Assumption A2 fails.

- F4 — β-Decay Observables: If future extensions predict angular correlations or spectral deviations excluded by experiment, the curvature-messenger interpretation is falsified.

Cross-Consistency Check: All tests feed into the central relation $\Delta\mu = \mu_0(\delta_W - \alpha)$. Violation of this equality falsifies the core structure of this appendix.

5.4 Summary Table: Key Parameters

Parameter	Value	Status	Source
π^2	9.8696...	[FUNDAMENTAL]	Geometry (Appx 7)
$j_{1,1}$	3.8317...	[DERIVED]	Bessel J_1 Zero
$\Delta\pi$	4.8124...	[DERIVED]	Stability Gap
ξ_*	4.8124...	[DERIVED]	AoC_1 (α)
μ_0	1707.36	[DERIVED]	$\Lambda^2_{\text{conf}}/(\xi_*\Delta\pi)$
α	0.00730...	[EXP/DERIVED]	CODATA / AoC_1
δ_W	0.00878...	[TARGET]	This Work ($\approx 1.2\alpha$)

Appendix Dependencies:

- Appx_7_AoC_01 (On_π_x): Provides π^2, Bessel spectrum, radial manifold.
- Appx_7_AoC_03 (On_137): 7b/7c (α): Provides ξ_*, α-derivation.
- Appx_7_AoC_04 (Proton): Provides μ_0, $\Gamma^{(p)}_{\text{bind}}$, container term.
- Appx 13 (Neutrinos): [FUTURE] Will provide δ_W from curvature coupling.
- Appx 12 (Forces): [FUTURE] Will provide δ_W from weak rung.